

AWS re/Start Task: Deploy Web Application Infrastructure with Bastion Host and RDS

Objective

In this task, you will build a secure and functional AWS infrastructure to host a simple web application. You will first create and configure the resources manually using the AWS Management Console. After verifying that everything works, you will document the steps and translate them into an AWS CLI script that replicates the entire infrastructure.

Architecture Overview

Your infrastructure will include:

- **A custom VPC**
 - **Two subnets:**
 - A **public subnet** for the Bastion Host and Web Server
 - A **private subnet** for the RDS database
 - **An Internet Gateway** for external access to the public subnet
 - **A Bastion Host** to securely access internal resources
 - **A Web Server** running a basic application
 - **An RDS instance** (MySQL or MariaDB) hosted in the private subnet
 - **Security Groups** to enforce proper access control between components
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Requirements

Networking

- Create a custom VPC (e.g., CIDR: **10.0.0.0/16**)
- Create two subnets:
 - **PublicSubnet** (e.g., **10.0.1.0/24**)
 - **PrivateSubnet** (e.g., **10.0.2.0/24**)
- Attach an Internet Gateway to the VPC
- Configure route tables:
 - Public subnet should route to the Internet Gateway
 - Private subnet should not have internet access
- Associate each subnet with the appropriate route table

EC2 Instances

- **Bastion Host:**
 - Located in the public subnet
 - Accessible via SSH only from your IP
- **Web Server:**
 - Located in the public subnet

- Accessible via HTTP from the internet
- SSH access only from the Bastion Host

RDS Database

- Deploy a MySQL or MariaDB RDS instance in the private subnet
- RDS should **only** allow inbound traffic on port 3306 from the Web Server

Security Groups

- Configure three security groups:
 1. Bastion: Allow SSH from your IP
 2. Web Server: Allow HTTP from anywhere and SSH from the Bastion only
 3. RDS: Allow MySQL/MariaDB traffic only from the Web Server
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Phase 1: Manual Deployment via AWS Console

1. Set up the VPC and subnets
 2. Attach Internet Gateway and configure route tables
 3. Create and assign appropriate security groups
 4. Launch:
 - A Bastion Host (Amazon Linux 2) in the public subnet
 - A Web Server (Amazon Linux 2) in the public subnet
 - An RDS instance (MySQL or MariaDB) in the private subnet
 5. Connect to the Web Server via the Bastion Host using SSH
 6. From the Web Server, test the connection to the RDS database
 7. (Optional) Install a simple application (e.g., PHP test page or WordPress) that connects to the database
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Phase 2: Recreate Infrastructure Using AWS CLI

Use Github to submit your work

- Create a new repository with a name like `aws-web-app-infrastructure`
- Add a README file with the task description
- Create a directory structure to organize your files
- Commit your changes regularly
- Push your repository to GitHub

AWS CLI Script

- Document all steps from the manual setup
- Translate each step into corresponding AWS CLI commands
- Organize the script into logical sections:
 - Networking
 - EC2 instance setup
 - Security groups
 - RDS setup

- Ensure the CLI script can deploy the infrastructure from scratch
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